

IoT in Smart Agriculture

Applications, Technologies and Challenges of the Internet of Things in Modern Farming

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1. Introduction

Agriculture must produce more food using less water, land and labour. Smart agriculture, also called precision agriculture, responds to this challenge by using data to guide every decision on the farm. The Internet of Things (IoT) is its technical backbone: a network of low-power sensors, controllers and communication links that collects field data in real time and turns it into action [1]. This report explains how IoT is applied in smart agriculture, the technologies behind it, its benefits, and the challenges that still limit adoption.

2. System Architecture

A typical IoT farming system is organised into three layers [3], [5]:

- **Perception layer:** sensors and actuators in the field, such as soil-moisture, pH, temperature and humidity sensors, plus relay-driven valves and pumps.
- **Network layer:** carries data from field nodes to a gateway and the cloud using links such as Wi-Fi, LoRaWAN, NB-IoT, Sigfox or satellite.
- **Application layer:** dashboards, mobile apps and analytics that give farmers irrigation scheduling, greenhouse management and supply-chain tracking [5].

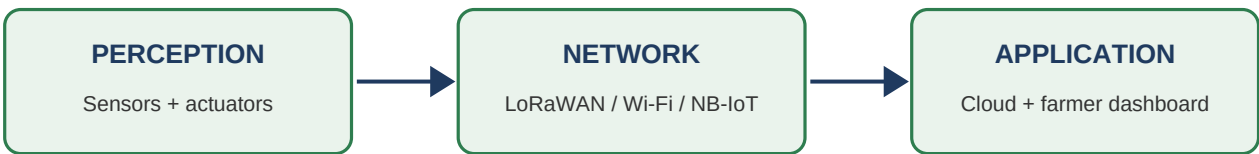


Figure 1. Three-layer architecture of an IoT-based smart farm.

Sensor	Parameter measured	Typical use
Soil moisture	Water content of the soil	Triggers irrigation
Soil pH / nutrients	Acidity and nutrient level	Fertiliser planning
Temperature and humidity	Air conditions around the crop	Disease risk, greenhouse control
Plant-stress sensors	Indicators of crop health	Early intervention
Weather inputs	Rainfall, wind, radiation	Irrigation forecasting

Table 1. Common sensors in smart agriculture [1], [5].

3. Key Applications

3.1 Smart irrigation

Irrigation is the most mature IoT use case. Field nodes read soil moisture and weather data, and a controller opens a valve when moisture drops below a threshold and closes it once the target is reached. In other words, the loop is sense, compare, act and report, repeated continuously without a person walking the field. Reviews of IoT irrigation describe systems built from sensors, actuators, nodes and wireless links that apply water only when and where it is needed [2]. Machine-learning models can also analyse wireless-sensor-network data to predict irrigation demand and automate control [5].

3.2 Soil and crop monitoring

Continuous readings of moisture, pH and plant stress allow targeted fertilisation and pest management instead of treating a whole field uniformly [1].

3.3 UAV-assisted farming

Drones combined with IoT networks cover large areas with minimal supervision, supporting automated operations on sustainable smart farms [3].

3.4 Greenhouse and environment control

Temperature, humidity and light sensors let a controller adjust fans, vents, misting and lighting automatically. Smart greenhouse management is one of the services delivered at the application layer of agricultural IoT platforms [5].

3.5 Data analytics and decision support

Sensor readings are stored in the cloud so that trends can be reviewed over a season rather than as isolated numbers. Dashboards show current field status, alerts warn of dry soil or abnormal conditions, and analytics models turn the history into recommendations [1], [5]. Processing simple threshold rules on the device itself (edge computing) keeps the system working even when the network link drops.

4. Connectivity for Remote Farms

Farms are large, remote and often outside cellular coverage, so the choice of communication technology is critical. LoRaWAN is widely studied because it combines low cost, long range and low power consumption, which suits battery-operated field sensors [4]. NB-IoT offers low latency, while Sigfox and LoRaWAN trade some latency for energy efficiency. Where coverage is poor, researchers have proposed meshed LoRaWAN gateways and satellite links [3].

5. Benefits

Data-driven decisions optimise resource use, raise efficiency and support sustainable practice [1]. Smart irrigation uses water more efficiently and reduces electricity consumption and labour costs, while sensors on plants and soil help deliver nutrients precisely and increase yields [4]. Remote monitoring also reduces the number of physical field visits, so farmers can manage several plots from a phone. At a larger scale, these gains contribute to global food security [1].

6. Challenges

- **Cost:** high initial investment; typical commercial irrigation sensors are expensive enough to put them out of reach of small farmers [1], [2].
- **Data and skills:** managing large data streams requires technical expertise that many farms lack [1].
- **Security and privacy:** connected devices and farm data must be protected [1].
- **Connectivity and power:** remote areas have weak networks and scarce power, so nodes must be low-power and resilient [1], [3].

7. A Low-Cost Prototype Perspective

The cost barrier can be reduced with hobbyist-grade electronics. An illustrative node uses an ESP32 or ESP8266 microcontroller, a soil-moisture sensor, a DHT temperature and humidity sensor, and a relay driving a small pump, publishing readings over MQTT to a free cloud dashboard. This reproduces the three-layer architecture in Figure 1 at a fraction of commercial cost and makes a practical student or hackathon project.

8. Future Trends

Precision agriculture is evolving from Agriculture 4.0 towards Agriculture 5.0 [1]. Combining IoT with machine learning, UAVs and low-cost long-range networks will make farm decisions more predictive and automated [3], [4], [5]. Future research is also expected to focus on better handling of data security and connectivity gaps, the two issues that most often slow deployment in practice [1], [3].

9. Conclusion

IoT turns farms into measurable, controllable systems. Smart irrigation, soil and crop monitoring and UAV support already improve resource efficiency, while LoRaWAN and satellite links extend coverage to remote land. Cost, skills, security and connectivity remain obstacles, but low-power electronics and affordable wireless networks are steadily lowering them, making IoT a practical route to sustainable and resilient food production.

References

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